SCALES, SYMMETRY AND WAVES ON ONE PAGE

Reductionism: parts and patterns

From cells and atoms to Fourier series, vibrating drums and the microwave sky.

1 / ASK WHAT COUNTS AS AN ELEMENT

Useful building blocks depend on the question and scale

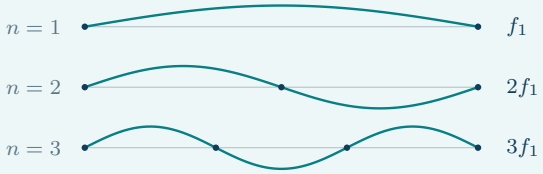
Reductionism studies parts and their interactions; a **normal mode** is a coordinated pattern of the whole.

Perspective	A useful decomposition	Relations still matter
Biology	Cell → organelles → molecules	Transport, reactions, feedback
Chemistry	Molecule → atoms → nuclei and electrons	Bonding and quantum interactions
Cosmology	Cosmic web → galaxies and diffuse matter	Gravity and expansion
Mathematics	Integers → primes; functions → basis functions	Factorisation or linear combination

2 / REDUCE A COMPLICATED SHAPE TO SIMPLE PATTERNS

Fourier series: the string's vocabulary

A string fixed at $x = 0, L$ has nodes at both ends. Its standing-wave shapes are $\sin(n\pi x/L)$.



Each mode oscillates at its own frequency:

$$f_n = \frac{nc}{2L}, \qquad u(x,t) = \sum_{n=1}^{\infty} q_n(t) \sin \frac{n\pi x}{L}.$$

For a fixed initial shape $g(x)$, its Fourier coefficients are

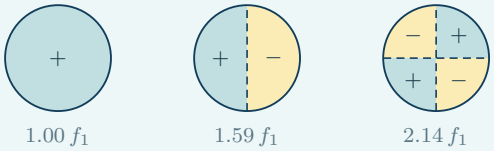
$$a_n = \frac{2}{L} \int_0^L g(x) \sin \frac{n\pi x}{L} \, \mathrm{d}x.$$

Orthogonality isolates each contribution. Initial shape and velocity fix the time coefficients $q_n(t)$.

3 / GIVE THE STRING A SECOND SPATIAL DIMENSION

A drum: geometry chooses the modes

A uniform membrane with a fixed rim has two-dimensional modes; internal nodal lines also stay still.



$1.00\, f_1$ $1.59\, f_1$ $2.14\, f_1$

Signs show displacement. These first three distinct frequencies are not integer harmonics.

$$-\nabla^2 \phi = k^2 \phi, \qquad f = \frac{ck}{2\pi}.$$

Isoperimetry and Faber–Krahn. For fixed area A, the disk D_A minimizes perimeter P and fundamental frequency:

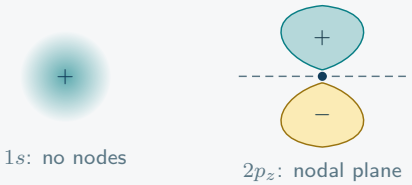
$$P^2 \geq 4\pi A, \qquad f_1(\Omega) \geq f_1(D_A)$$

4 / THE ATOM HAS ITS OWN ALLOWED SPATIAL PATTERNS

Atomic orbitals are quantum modes

Hydrogen's electron is bound by the nucleus's electric potential V. Its allowed states solve the Schrödinger equation:

$$\left[-\frac{\hbar^2}{2m_e} \nabla^2 + V(r) \right] \psi = E\psi.$$



$1s$: no nodes $2p_z$: nodal plane

Schematic shapes; colour marks the sign of a real ψ. The probability density is $|\psi|^2$.

Hydrogen separates into radial and angular modes:

$$\psi_{n\ell m}(r,\theta,\varphi) = R_{n\ell}(r) Y_{\ell m}(\theta,\varphi).$$

An energy eigenstate has a time-independent probability density. Transitions give spectral lines $hf = |E_b - E_a|$; many-electron atoms add electron interactions.

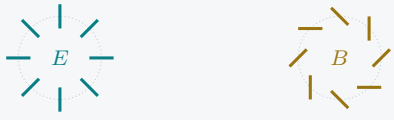
5 / FROM CATALOGUES OF OBJECTS TO MODES OF A FIELD

The microwave sky has a spectrum

Galaxy catalogues describe objects. Cosmic microwave background radiation (CMBR) supplies a field on the sky. Its temperature fluctuations have a spherical Fourier expansion:

$$\Delta T(\theta,\varphi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\theta,\varphi).$$

Hydrogen uses the same $Y_{\ell m}$. CMB acoustic peaks record oscillations in the early plasma of light and matter.



even parity odd parity

Polarization separates into **E and B modes** with even and odd parity under reflection (schematic above).

Lensing converts E into B; primordial gravitational waves would also generate B. Separating lensing and foregrounds is essential when testing early-Universe models.

The shared idea. Choose variables and constraints, find the allowed patterns, and determine their weights. Mode decomposition is a change of description within a linear model; an explanation also needs interactions, organisation and evidence.

Conventions. c is the classical wave speed; drums have the same uniform tension and mass per area, with fixed rims. Atomic formulas use the nonrelativistic fixed-nucleus hydrogen model.

Sources. MIT: Fourier and standing waves; MIT: orbitals; Daners: Faber–Krahn; WMAP: angular spectrum; Hanson et al.: lensing B modes.

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